

Review Session

1. Prior Knowledge: Creating/Instantiating/Extending Classes

2. Exception Handling

- a. Try-catch block
- b. Throw statement
- c. Declare rule (throws clause and no try-catch block)
- d. Catch rule (try-catch block and no throws clause)
- e. Creating Exception Classes
- f. Checked/Unchecked Exceptions

3. File IO

- a. Accessing File properties (class File)
- b. Opening/Closing files for reading (File and Scanner classes)
- c. Reading from files (Scanner methods)
- d. Opening/Closing files for writing (File and PrintWriter classes)
- e. Writing to files (PrintWriter methods)
- f. Checked exception FileNotFoundException thrown by the constructors in the classes Scanner and PrintWriter

4. Regular Expressions

- a. The methods **matches()** and **split()** from the class String
- b. Simple regular expressions

5. Interfaces

- a. Classes implement interfaces (not extend) by defining the methods from the interface
- b. Cannot be instantiated
- c. Examples: Comparable (**compareTo**) and Cloneable (**clone: shallow or deep copy**)
- d. Can be used as type to hold instances of the sub classes
- e. **Upcasting**: assigning an instance of the sub type to a variable of the super type - done automatically (polymorphism)
- f. **Downcasting**: assigning an instance of the super type to a variable of the subtype - must be explicitly done and safely through the instanceof operator
- g. **instanceof** operator